

Position, Not Capability: How Stock Markets Price Large Language Model Releases

Zhuo Chen^{a,*}, Ruishuang Lu^a

^a*Center for Economic Research, Shandong University, Jinan, China*

Abstract

Do stock markets price the capability of large language models, or the economic position of firms around a release? Using 124 day-verified model releases (2022–2026), benchmark capability scores, and double-blind-coded ecosystem positions for Nasdaq-100 firms, we find capability is not positively priced: the slope is zero pooled, significantly negative among closed-source flagships, and zero among unrelated firms. Position is priced: hardware suppliers and rival developers earn 1.6–1.7 percentage points over twenty days, with immediate abnormal volume, while firms deploying AI in conventional businesses show precisely zero price and volume response. Capability matters only as an amplifier of the hardware premium.

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*Corresponding author.

Email address: zhuochen@sdu.edu.cn (Zhuo Chen)

1. Introduction

Large language model (LLM) releases have become routine, salient information events—124 benchmark-qualified releases in our 2022–2026 sample. A natural reading, implicit in exposure-based studies of AI and asset prices (Eisfeldt et al., 2023; Babina et al., 2024), is that markets reward technological capability: the more capable the model, the larger the revaluation of exposed firms. This letter tests that reading directly and rejects it. Markets price *where a firm sits* in the AI production chain, not *how capable* the released model is.

We combine three measurement inputs that are, to our knowledge, new in combination. First, release dates verified to the day against first-party sources (vendor blogs, release notes, archived first captures) for every event. Second, a direct measure of model capability—Artificial Analysis benchmark scores—rather than a proxy. Third, ecosystem positions (hardware supplier, cloud platform, integrator, deployer, competitor, investor, owner) assigned to each (firm, developer) pair by two independent coders under a fixed codebook (agreement 99.0%, Cohen’s κ 0.84–1.00). The firm universe is the Nasdaq-100, so selection is exogenous to AI exposure and the thirty-one constituents with no AI relationship provide a within-event control group.

Three results follow. First, capability is not positively priced. The pooled slope of twenty-day cumulative abnormal returns (CARs) on capability is zero; among closed-source releases—where capability rents would be most appropriable (Teece, 1986)—it is significantly *negative* (-0.8 pp per standard deviation of capability, $p = 0.02$); among firms with no ecosystem relationship it is an exact zero, as a placebo requires. The negative flagship slope is

an anticipation gradient: the most capable releases are the most rumored, so their information reaches prices before the event window. Second, position is priced. Hardware suppliers earn 1.66 pp and rival model developers 1.61 pp over $[0, +20]$ ($p \approx 0.01$), with abnormal trading volume on the announcement day itself; deployers—firms using AI inside conventional businesses—show a CAR of -0.07 pp (SE 0.25 pp) and no abnormal volume: a genuine zero, not offsetting disagreement. Third, the two dimensions interact: capability carries no average price but amplifies the hardware premium (interaction $p = 0.088$), exactly as derived-demand logic implies—a more capable model is bigger news for compute suppliers, not for the average “exposed” firm.

The letter’s contribution is to unbundle “AI exposure.” Exposure rankings conflate firms on opposite sides of the same shock; once positions are separated, the pricing loads on the compute-complementarity channel and on industry information spillovers to rivals (Lang and Stulz, 1992), while the displacement channel that dominates public commentary is priced at zero. A companion full-length paper documents when this pricing rule formed (2025, with the reasoning paradigm) and shows that two opposite headline results in an earlier hand-collected vintage of the data were artifacts of compound measurement error; here we hold the reproducible pipeline fixed and focus on the cross-sectional question.

2. Data

Events intersect the AI Timeline chronology with Artificial Analysis (AA) benchmark coverage, which screens out releases too minor to be benchmarked; identity matches and all manual decisions are recorded in auditable

decision tables. Of 134 candidate events, 125 have day-verified dates and 124 enter the main sample (3 events from 2022, 8 from 2023, 45 from 2024, 59 from 2025, 9 from 2026); 87 are language-model releases with AA intelligence scores, 47 are open-weight. Capability is the AA intelligence index of the event’s flagship model (SD 12.7 in the estimation sample).

For each of the 124 events and 101 Nasdaq-100 firms we compute market-model CARs (S&P 500 proxy, estimation window $[-200, -10]$, minimum 120 observations) over $[0, +w]$, $w \leq 20$, and standardized abnormal log volume; three-factor CARs serve as robustness. Firm controls (log assets, book-to-market with a negative-equity flag, momentum, volatility) are matched under a strict 45-day as-of rule. Positions at the (firm, developer) level are aggregated to the event level by union over the event’s developers. Standard errors are clustered by event (CR2) throughout; specifications with event fixed effects identify from within-event variation only.

3. Results

3.1. *Capability is not priced*

Table 1 regresses $\text{CAR}[0, +20]$ on the capability score with firm controls and year fixed effects. The pooled slope is -0.0001 per index point ($p = 0.51$). Among the 50 closed-source releases the slope is -0.0006 ($p = 0.024$; -1.3 pp per SD under the three-factor benchmark, $p < 0.001$); among the 37 open-weight releases it is 0.0000 ($p = 0.93$). Among unrelated firms the slope is zero ($p = 0.80$)—capability scores do not predict returns where no economic channel exists, so the closed-source negative is not a mechanical artifact. Replacing the level with a surprise measure (the score’s distance

Table 1: Model capability and abnormal returns

	(1) All	(2) Closed	(3) Open	(4) Placebo (unrelated firms)	(5) Interaction
AA intelligence	-0.0001 (0.0002)	-0.0006** (0.0002)	0.0000 (0.0003)	-0.0001 (0.0004)	-0.0003 (0.0002)
× Upstream hardware					0.0010* (0.0006)
Upstream hardware					0.0207*** (0.0078)
Events	87	50	37	87	87
Observations	8,562	4,917	3,645	2,784	8,562

Notes: Dependent variable is the market-model CAR[0, +20]. All columns include firm controls (log assets, book-to-market with negative-equity flag, volatility, momentum) and year fixed effects; capability is centered in column (5). Standard errors clustered by event (CR2). *, **, *** denote 10%, 5%, 1%.

above the pre-event frontier maximum) yields the same pattern, because benchmark surprise still correlates with flagship status: the releases that move benchmarks most are the ones markets see coming. Column (5) shows the interaction that survives: capability amplifies the hardware premium (+0.0010 per index point, $p = 0.088$; the hardware main effect at mean capability is +2.07 pp, $p = 0.010$).

3.2. Position is priced

Table 2 regresses the same outcome on the seven position indicators. Hardware suppliers earn 1.66 pp over unrelated constituents ($p = 0.012$;

1.79 pp under three factors; 1.54 pp with event fixed effects), and rival developers earn 1.61 pp—a positive competitor spillover indicating that releases convey industry-level growth information that dominates competitive displacement (Lang and Stulz, 1992; Chen et al., 2005). The deployer coefficient is -0.07 pp with a 95% confidence interval of $[-0.56, +0.42]$ pp: the market does not price LLM releases as bad news for firms that merely use AI. The premium is not a single-stock artifact—dropping any one hardware firm leaves it between 1.2 and 1.6 pp, and excluding NVIDIA gives 1.5 pp ($p = 0.023$).

Figure 1 traces the accumulation. The hardware premium is absent at the announcement close, significant from $[0, +10]$ in the full sample—and from $[0, +2]$ in the 2025–2026 subsample, where it reaches 4.3 pp by day twenty—while the deployer line never leaves zero. Controlling for each firm’s pre-event CAR leaves the estimates unchanged, so the accumulation is not pre-existing drift.

3.3. Volume separates attention from irrelevance

Abnormal volume completes the taxonomy (Table 3). Hardware suppliers trade 0.12 standard deviations above normal on the announcement day ($p < 0.001$)—attention is immediate even where same-day price response is nil, and directional consensus forms over the following weeks. Deployers show no abnormal volume at any horizon ($p = 0.74$ at $[0, +1]$): their zero price response reflects irrelevance, not offsetting disagreement. Cloud platforms are the third fate: the largest abnormal volume in the panel (0.29 SD at $[0, +1]$, and already 0.20 SD in the pre-event window) with no directional price response—the signature of investors interpreting the same signal with

Table 2: Ecosystem position and abnormal returns

Dependent variable:	(1) $CAR^{MM}[0, 20]$	(2) $CAR^{FF3}[0, 20]$	(3) $CAR^{MM}[0, 20]$
Upstream hardware	0.0166** (0.0065)	0.0179*** (0.0065)	0.0154** (0.0066)
Upstream cloud	-0.0011 (0.0078)	0.0020 (0.0074)	-0.0018 (0.0076)
Downstream integrator	-0.0051 (0.0049)	-0.0041 (0.0048)	-0.0053 (0.0050)
Downstream deployer	-0.0007 (0.0025)	0.0002 (0.0024)	-0.0011 (0.0025)
Competitor	0.0161** (0.0064)	0.0159*** (0.0058)	0.0158** (0.0064)
Investor, owner	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes
Year fixed effects	Yes	Yes	—
Event fixed effects	—	—	Yes
Observations	12,112	12,112	12,112
Events	124	124	124

Notes: Investor and owner/publisher indicators included (positive, insignificant; few clusters).

Standard errors clustered by event (CR2). *, **, *** denote 10%, 5%, 1%.

opposite priors (Kandel and Pearson, 1995), natural for firms for which a frontier release is simultaneously compute-rental good news and own-model bad news.

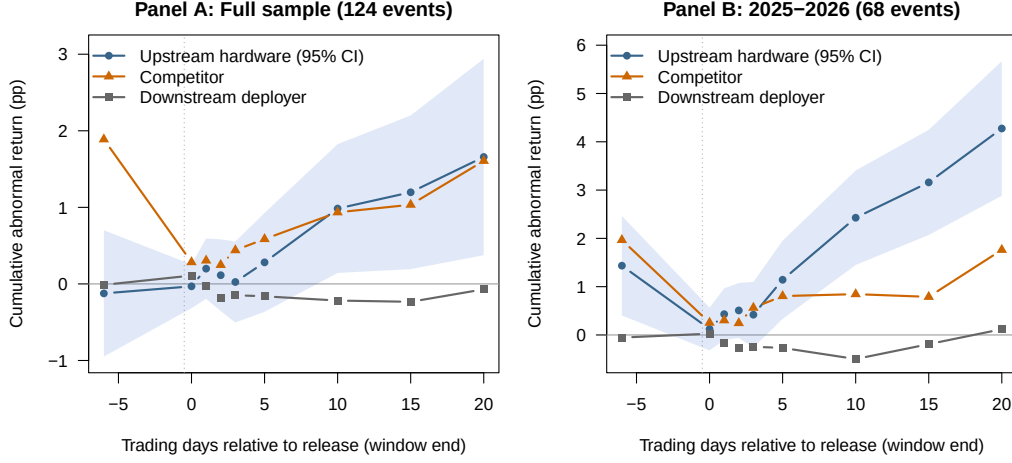


Figure 1: Cumulative abnormal returns by ecosystem position and accumulation window. Shading is the 95% band for the hardware coefficient (CR2, by event). Left panel: full sample; right panel: 2025–2026.

3.4. Robustness

The position premia are significant in 16–17 of 20 specifications spanning benchmarks (S&P 500, Nasdaq-100, SOX, three-factor), windows, winsorization, date-confidence subsamples, multi-component-event exclusion, and event fixed effects; the deployer zero and a null hardware $\times$ open-weight interaction ($p = 0.90$) hold in all twenty. Announcement timestamps recovered for 55 events show only two after-hours releases; shifting all events by one trading day leaves the results unchanged. The capability results are insensitive to using capability surprise, to three-factor benchmarks, and to restricting to firms with any AI relationship.

Table 3: Abnormal trading volume

Window:	$[-10, -2]$	$[0, +1]$
Upstream hardware	0.033 (0.026)	0.121*** (0.036)
Upstream cloud	0.204*** (0.041)	0.292*** (0.049)
Downstream deployer	-0.019 (0.015)	0.006 (0.019)
Competitor	-0.037 (0.029)	-0.017 (0.043)
Event fixed effects	Yes	Yes

Notes: Standardized abnormal log volume (deviation from the $[-200, -10]$ mean in estimation-window SDs, averaged over the window). Firm controls included; SEs clustered by event (CR2).

4. Conclusion

Around LLM releases, equity markets do not pay for capability—they pay for position. Benchmark scores carry no positive premium and a negative one among anticipated flagships; hardware suppliers and rival developers revalue by 1.6–1.7 pp while AI-deploying firms show precisely zero price and volume response; and capability matters only by scaling the hardware premium. For research that ranks firms by AI exposure, the lesson is that exposure conflates positions with opposite pricing; for the public debate, four years of releases contain no evidence that markets price AI displacement on announcement.

Data availability

The event pipeline, coding decision tables, and estimation code are available from the authors.

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